

Laplace AR(1) under Ornstein–Uhlenbeck Corruption

A detailed derivation note for the score, the residual representation, the Fourier picture, and the curvature field

This note is written to be editable: every symbol is defined, every transformation is shown, and the key formulas are isolated so they can be modified easily.

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1 Roadmap

We study the scalar AR(1) model with Laplace innovation,

$$A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad A_1 = \alpha A_0 + \varepsilon, \quad \varepsilon \sim \text{Laplace}(0, b), \quad (1)$$

with independent coordinates at the clean level:

$$\varepsilon \perp A_0. \quad (2)$$

The observed noisy variables are generated through the Ornstein–Uhlenbeck (OU) channel

$$X_k = \beta A_k + \sqrt{\Delta} Z_k, \quad \beta = e^{-t}, \quad \Delta = 1 - e^{-2t} = 1 - \beta^2, \quad Z_0, Z_1 \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1). \quad (3)$$

We want explicit formulas for

$$p_t(x_0, x_1), \quad \mathcal{S}(x, t) = \nabla_x \log p_t(x), \quad \mathcal{H}_t(x) = -\nabla_x^2 \log p_t(x). \quad (4)$$

The central message is the following.

For $K = 1$, the whole two-dimensional problem collapses to one Gaussian site variable x_0 and one non-Gaussian residual variable

$$r = x_1 - \nu - \rho x_0.$$

The joint law factors as

$$p_t(x_0, x_1) = \text{Gaussian in } x_0 \times \text{Normal-Laplace in } r,$$

and the full Hessian field is controlled by a single scalar function

$$\kappa_t(r) = -\frac{d^2}{dr^2} \log h_t(r),$$

where h_t is the density of a Gaussian plus a scaled Laplace random variable.

Everything that looks two-dimensional in the heatmaps is therefore the image of a one-dimensional residual geometry transported along straight lines $r = \text{constant}$.

2 Model, notation, and baseline densities

2.1 Clean density

The Laplace density with location 0 and scale $b > 0$ is

$$f_\varepsilon(u) = \frac{1}{2b} \exp\left(-\frac{|u|}{b}\right). \quad (5)$$

Therefore the clean joint density of (A_0, A_1) is

$$p_0(a_0, a_1) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(a_0 - \mu_0)^2}{2\sigma_0^2}\right) \cdot \frac{1}{2b} \exp\left(-\frac{|a_1 - \alpha a_0|}{b}\right). \quad (6)$$

This formula already shows the clean geometry:

- the *root channel* a_0 is Gaussian;
- the *innovation channel* $a_1 - \alpha a_0$ is Laplace;
- the non-Gaussianity is concentrated on the bond variable $a_1 - \alpha a_0$.

2.2 Noisy density as a Gaussian channel mixture

Conditionally on $A = (A_0, A_1)$, the noisy law is Gaussian with independent coordinates:

$$g_t(x|a) = \frac{1}{2\pi\Delta} \exp\left(-\frac{(x_0 - \beta a_0)^2 + (x_1 - \beta a_1)^2}{2\Delta}\right). \quad (7)$$

Hence the noisy density is

$$p_t(x_0, x_1) = \int_{\mathbb{R}^2} g_t(x|a) p_0(a_0, a_1) da_0 da_1. \quad (8)$$

Because the clean law is Markov and the channel is coordinatewise conditionally independent, the noisy law preserves a one-step factorization:

$$p_t(x_0, x_1) = p_t(x_0) p_t(x_1 | x_0). \quad (9)$$

Therefore the score is automatically local in the Markov sense:

$$\mathcal{S}(x, t) = \begin{pmatrix} \partial_{x_0} \log p_t(x_0) + \partial_{x_0} \log p_t(x_1 | x_0) \\ \partial_{x_1} \log p_t(x_1 | x_0) \end{pmatrix}. \quad (10)$$

Equation (10) is the first exact locality statement. The second component depends only on the single edge potential $p_t(x_1 | x_0)$; the first component is a sum of one site term and one edge correction.

3 The x_0 -marginal and the posterior law of $A_0 | X_0$

Since

$$X_0 = \beta A_0 + \sqrt{\Delta} Z_0, \quad A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad Z_0 \sim \mathcal{N}(0, 1), \quad (11)$$

we immediately obtain

$$X_0 \sim \mathcal{N}(\beta\mu_0, v), \quad v := \beta^2\sigma_0^2 + \Delta. \quad (12)$$

Thus

$$p_t(x_0) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x_0 - \beta\mu_0)^2}{2v}\right), \quad (13)$$

and the corresponding marginal score is

$$\partial_{x_0} \log p_t(x_0) = -\frac{x_0 - \beta\mu_0}{v}. \quad (14)$$

Now compute the posterior law of $A_0 | X_0 = x_0$. Since everything is jointly Gaussian, it is again Gaussian. The standard conditioning formulas give

$$A_0 | X_0 = x_0 \sim \mathcal{N}(m(x_0), s^2), \quad (15)$$

with

$$m(x_0) = \mu_0 + \frac{\text{Cov}(A_0, X_0)}{\text{Var}(X_0)}(x_0 - \beta\mu_0), \quad s^2 = \text{Var}(A_0) - \frac{\text{Cov}(A_0, X_0)^2}{\text{Var}(X_0)}. \quad (16)$$

Because

$$\text{Cov}(A_0, X_0) = \text{Cov}(A_0, \beta A_0 + \sqrt{\Delta} Z_0) = \beta\sigma_0^2, \quad (17)$$

we obtain

$$m(x_0) = \mu_0 + \frac{\beta\sigma_0^2}{v}(x_0 - \beta\mu_0) = \frac{\Delta\mu_0 + \beta\sigma_0^2 x_0}{v}, \quad (18)$$

and

$$s^2 = \sigma_0^2 - \frac{\beta^2\sigma_0^4}{v} = \frac{\sigma_0^2\Delta}{v}. \quad (19)$$

These two quantities are fundamental. The noisy root variable X_0 carries all available information about A_0 , and all remaining uncertainty is contained in the posterior variance s^2 .

4 Conditional law of $X_1 | X_0 = x_0$: reduction to one residual variable

The next task is to compute $p_t(x_1 | x_0)$. Start from

$$X_1 = \beta A_1 + \sqrt{\Delta} Z_1 = \beta(\alpha A_0 + \varepsilon) + \sqrt{\Delta} Z_1. \quad (20)$$

Insert the decomposition

$$A_0 = m(x_0) + \xi, \quad \xi | X_0 = x_0 \sim \mathcal{N}(0, s^2), \quad \xi \perp (X_0, \varepsilon, Z_1). \quad (21)$$

Then

$$X_1 = \beta \alpha m(x_0) + \beta \alpha \xi + \beta \varepsilon + \sqrt{\Delta} Z_1 \quad (22)$$

$$= \beta \alpha m(x_0) + \beta \varepsilon + G_t, \quad (23)$$

where

$$G_t := \beta \alpha \xi + \sqrt{\Delta} Z_1 \sim \mathcal{N}(0, \tau^2), \quad (24)$$

with variance

$$\tau^2 = \beta^2 \alpha^2 s^2 + \Delta = \Delta + \alpha^2 \beta^2 \frac{\sigma_0^2 \Delta}{v}. \quad (25)$$

We now isolate the x_0 -dependence of the mean term:

$$\beta \alpha m(x_0) = \nu + \rho x_0, \quad (26)$$

with

$$\rho := \alpha \frac{\beta^2 \sigma_0^2}{v}, \quad \nu := \alpha \beta \frac{\Delta \mu_0}{v}. \quad (27)$$

Indeed, substituting (18) gives

$$\beta \alpha m(x_0) = \beta \alpha \frac{\Delta \mu_0 + \beta \sigma_0^2 x_0}{v} = \alpha \beta \frac{\Delta \mu_0}{v} + \alpha \frac{\beta^2 \sigma_0^2}{v} x_0. \quad (28)$$

Therefore

$$X_1 | X_0 = x_0 = \nu + \rho x_0 + R_t, \quad (29)$$

where the residual variable is

$$R_t := \beta \varepsilon + G_t, \quad G_t \sim \mathcal{N}(0, \tau^2), \quad R_t \perp X_0. \quad (30)$$

This is the main structural reduction. The entire non-Gaussian problem is now encoded in the one-dimensional random variable R_t , which is a Gaussian plus a scaled Laplace variable.

4.1 Exact factorized form of the joint density

Let h_t denote the density of R_t . Then (29) implies

$$p_t(x_1 | x_0) = h_t(x_1 - \nu - \rho x_0). \quad (31)$$

Combining this with (13) yields the exact joint density

$$p_t(x_0, x_1) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x_0 - \beta\mu_0)^2}{2v}\right) h_t(r), \quad r := x_1 - \nu - \rho x_0. \quad (32)$$

From this point onward, every derivative can be computed by the chain rule from a one-dimensional function h_t .

5 Derivation of the Normal–Laplace density h_t

By construction,

$$h_t(r) = \int_{\mathbb{R}} \frac{1}{2\tilde{b}} \exp\left(-\frac{|u|}{\tilde{b}}\right) \cdot \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{(r-u)^2}{2\tau^2}\right) du, \quad (33)$$

where

$$\tilde{b} := \beta b. \quad (34)$$

We now compute this integral explicitly. Split the real line into positive and negative parts:

$$h_t(r) = \frac{1}{2\tilde{b}\sqrt{2\pi\tau^2}} \left[\int_0^\infty \exp\left(-\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2}\right) du + \int_{-\infty}^0 \exp\left(\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2}\right) du \right]. \quad (35)$$

5.1 First integral: $u \geq 0$

Consider

$$I_+(r) := \int_0^\infty \exp\left(-\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2}\right) du. \quad (36)$$

Expand the exponent and complete the square in u :

$$-\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2} = -\frac{u}{\tilde{b}} - \frac{u^2 - 2ru + r^2}{2\tau^2} \quad (37)$$

$$= -\frac{u^2}{2\tau^2} + \left(\frac{r}{\tau^2} - \frac{1}{\tilde{b}}\right)u - \frac{r^2}{2\tau^2}. \quad (38)$$

Now write

$$\frac{r}{\tau^2} - \frac{1}{\tilde{b}} = \frac{r - \tau^2/\tilde{b}}{\tau^2}, \quad (39)$$

so

$$-\frac{u^2}{2\tau^2} + \frac{r - \tau^2/\tilde{b}}{\tau^2}u = -\frac{1}{2\tau^2} \left[u^2 - 2(r - \tau^2/\tilde{b})u \right] \quad (40)$$

$$= -\frac{1}{2\tau^2} \left[(u - r + \tau^2/\tilde{b})^2 - (r - \tau^2/\tilde{b})^2 \right]. \quad (41)$$

Hence

$$-\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2} = -\frac{(u-r+\tau^2/\tilde{b})^2}{2\tau^2} + \frac{(r-\tau^2/\tilde{b})^2 - r^2}{2\tau^2} \quad (42)$$

$$= -\frac{(u-r+\tau^2/\tilde{b})^2}{2\tau^2} - \frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}. \quad (43)$$

Therefore

$$I_+(r) = \exp\left(-\frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}\right) \int_0^\infty \exp\left(-\frac{(u-r+\tau^2/\tilde{b})^2}{2\tau^2}\right) du. \quad (44)$$

Now use the substitution

$$z = \frac{u-r+\tau^2/\tilde{b}}{\tau}, \quad du = \tau dz, \quad u=0 \implies z = -\frac{r}{\tau} + \frac{\tau}{\tilde{b}}. \quad (45)$$

Then

$$I_+(r) = \exp\left(-\frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}\right) \tau \int_{-r/\tau+\tau/\tilde{b}}^\infty \exp\left(-\frac{z^2}{2}\right) dz \quad (46)$$

$$= \exp\left(-\frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}\right) \tau \sqrt{2\pi} \Phi\left(\frac{r}{\tau} - \frac{\tau}{\tilde{b}}\right). \quad (47)$$

5.2 Second integral: $u \leq 0$

Similarly define

$$I_-(r) := \int_{-\infty}^0 \exp\left(\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2}\right) du. \quad (48)$$

Complete the square again:

$$\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2} = -\frac{u^2}{2\tau^2} + \left(\frac{r}{\tau^2} + \frac{1}{\tilde{b}}\right)u - \frac{r^2}{2\tau^2} \quad (49)$$

$$= -\frac{(u-r-\tau^2/\tilde{b})^2}{2\tau^2} + \frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}. \quad (50)$$

Therefore

$$I_-(r) = \exp\left(\frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}\right) \int_{-\infty}^0 \exp\left(-\frac{(u-r-\tau^2/\tilde{b})^2}{2\tau^2}\right) du. \quad (51)$$

With the substitution

$$z = \frac{u-r-\tau^2/\tilde{b}}{\tau}, \quad u=0 \implies z = -\frac{r}{\tau} - \frac{\tau}{\tilde{b}}, \quad (52)$$

we obtain

$$I_-(r) = \exp\left(\frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}\right) \tau \int_{-\infty}^{-r/\tau-\tau/\tilde{b}} \exp\left(-\frac{z^2}{2}\right) dz \quad (53)$$

$$= \exp\left(\frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}\right) \tau \sqrt{2\pi} \Phi\left(-\frac{r}{\tau} - \frac{\tau}{\tilde{b}}\right). \quad (54)$$

5.3 Closed form

Insert the two formulas into (35). The factors $\tau\sqrt{2\pi}$ cancel the Gaussian normalization, giving

$$h_t(r) = \frac{\exp(a^2/2)}{2\tilde{b}} \left[\exp\left(-\frac{r}{\tilde{b}}\right) \Phi\left(\frac{r}{\tau} - a\right) + \exp\left(\frac{r}{\tilde{b}}\right) \Phi\left(-\frac{r}{\tau} - a\right) \right], \quad a := \frac{\tau}{\tilde{b}}. \quad (55)$$

This is the exact density of a Gaussian plus a scaled Laplace variable, often called the *Normal-Laplace* law.

6 Free energy form of the noisy law

Using (32), the log-density is

$$\log p_t(x_0, x_1) = -\frac{(x_0 - \beta\mu_0)^2}{2v} + \log h_t(r) + \text{const}. \quad (56)$$

Hence the negative log-density (free energy) is

$$F_t(x_0, x_1) := -\log p_t(x_0, x_1) = \frac{(x_0 - \beta\mu_0)^2}{2v} + U_t(r) + \text{const}, \quad U_t(r) := -\log h_t(r). \quad (57)$$

This decomposition is very important conceptually:

- the term $\frac{(x_0 - \beta\mu_0)^2}{2v}$ is a quadratic site energy;
- the term $U_t(r)$ is a bond energy depending on one residual variable only;
- Gaussianity is equivalent to U_t being quadratic;
- non-Gaussianity appears exactly as non-quadratic residual geometry.

7 Score: exact formulas and nonlinear structure

Define the residual score

$$\psi_t(r) := \frac{d}{dr} \log h_t(r) = \frac{h'_t(r)}{h_t(r)}. \quad (58)$$

Then from (56) and the chain rule

$$\partial_{x_1} r = 1, \quad \partial_{x_0} r = -\rho. \quad (59)$$

Therefore

$$\mathcal{S}(x, t) = \begin{pmatrix} -\frac{x_0 - \beta\mu_0}{v} - \rho \psi_t(r) \\ \psi_t(r) \end{pmatrix}. \quad (60)$$

This formula cleanly separates the contributions:

- $-(x_0 - \beta\mu_0)/v$: Gaussian root contribution;
- $\psi_t(r)$: nonlinear innovation contribution;

- $-\rho\psi_t(r)$: feedback of the bond onto the root score.

7.1 Explicit formula for ψ_t

Introduce

$$L_+(r) := \exp\left(-\frac{r}{\tilde{b}}\right) \Phi\left(\frac{r}{\tau} - a\right), \quad L_-(r) := \exp\left(\frac{r}{\tilde{b}}\right) \Phi\left(-\frac{r}{\tau} - a\right), \quad a = \frac{\tau}{\tilde{b}}. \quad (61)$$

Then (55) becomes

$$h_t(r) = \frac{\exp(a^2/2)}{2\tilde{b}} (L_+(r) + L_-(r)). \quad (62)$$

Differentiate term by term:

$$L'_+(r) = -\frac{1}{\tilde{b}} L_+(r) + \frac{1}{\tau} \exp\left(-\frac{r}{\tilde{b}}\right) \varphi\left(\frac{r}{\tau} - a\right), \quad (63)$$

$$L'_-(r) = \frac{1}{\tilde{b}} L_-(r) - \frac{1}{\tau} \exp\left(\frac{r}{\tilde{b}}\right) \varphi\left(\frac{r}{\tau} + a\right). \quad (64)$$

Now use the identity

$$\exp\left(-\frac{r}{\tilde{b}}\right) \varphi\left(\frac{r}{\tau} - a\right) = \exp\left(\frac{r}{\tilde{b}}\right) \varphi\left(\frac{r}{\tau} + a\right) = \exp\left(-\frac{a^2}{2}\right) \varphi\left(\frac{r}{\tau}\right). \quad (65)$$

Hence the Gaussian derivative pieces cancel in $L'_+ + L'_-$, and we obtain

$$h'_t(r) = \frac{\exp(a^2/2)}{2\tilde{b}^2} (L_-(r) - L_+(r)). \quad (66)$$

Dividing by $h_t(r)$ gives the compact exact expression

$$\boxed{\psi_t(r) = \frac{1}{\tilde{b}} \frac{L_-(r) - L_+(r)}{L_+(r) + L_-(r)}}. \quad (67)$$

7.2 Tail behavior of the score

Because $L_+(r)$ dominates for large positive r and $L_-(r)$ dominates for large negative r , we obtain

$$\psi_t(r) \rightarrow -\frac{1}{\tilde{b}} \quad (r \rightarrow +\infty), \quad \psi_t(r) \rightarrow +\frac{1}{\tilde{b}} \quad (r \rightarrow -\infty). \quad (68)$$

Therefore the score saturates in the tails. This is a sharp contrast with the Gaussian case, where the score is linear and therefore diverges in magnitude as $|r| \rightarrow \infty$.

8 Curvature field: exact matrix structure

Define the scalar residual curvature

$$\kappa_t(r) := -\psi'_t(r) = -\frac{d^2}{dr^2} \log h_t(r). \quad (69)$$

Differentiate the score (60) again. Since

$$\partial_{x_1} \psi_t(r) = \psi'_t(r), \quad \partial_{x_0} \psi_t(r) = -\rho \psi'_t(r), \quad (70)$$

we get

$$\partial_{x_1} S_1 = \psi'_t(r), \quad (71)$$

$$\partial_{x_0} S_1 = -\rho \psi'_t(r), \quad (72)$$

$$\partial_{x_1} S_0 = -\rho \psi'_t(r), \quad (73)$$

$$\partial_{x_0} S_0 = -\frac{1}{v} + \rho^2 \psi'_t(r). \quad (74)$$

Therefore

$$\mathcal{H}_t(x) = -\nabla_x^2 \log p_t(x) = \begin{pmatrix} \frac{1}{v} & 0 \\ 0 & 0 \end{pmatrix} + \kappa_t(r) \begin{pmatrix} \rho^2 & -\rho \\ -\rho & 1 \end{pmatrix}. \quad (75)$$

Componentwise,

$$H_{11}(x) = \kappa_t(r), \quad H_{01}(x) = -\rho \kappa_t(r), \quad H_{00}(x) = \frac{1}{v} + \rho^2 \kappa_t(r). \quad (76)$$

This tells us something much stronger than mere state dependence: the full two-dimensional curvature field lives on a one-dimensional manifold of matrices. Indeed,

$$H_{01}(x) = -\rho H_{11}(x), \quad H_{00}(x) - \frac{1}{v} = \rho^2 H_{11}(x). \quad (77)$$

So once H_{11} is known, all other entries are fixed.

8.1 Alternative formula for κ_t

Since

$$\kappa_t(r) = -\left(\frac{h_t''(r)}{h_t(r)} - \frac{h_t'(r)^2}{h_t(r)^2} \right) = \psi_t(r)^2 - \frac{h_t''(r)}{h_t(r)}, \quad (78)$$

it is useful to compute h_t'' . From the linear differential equation derived later in (107), one gets

$$h_t''(r) = \frac{h_t(r) - \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{r^2}{2\tau^2}\right)}{\tilde{b}^2}. \quad (79)$$

Therefore

$$\kappa_t(r) = \psi_t(r)^2 - \frac{1}{\tilde{b}^2} + \frac{1}{\tilde{b}^2} \frac{\frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{r^2}{2\tau^2}\right)}{h_t(r)}. \quad (80)$$

This formula is useful for asymptotics and for numerical evaluation.

9 Relative variables and the innovation channel

A natural variable is

$$y := x_1 - \alpha x_0. \quad (81)$$

At the clean level,

$$A_1 - \alpha A_0 = \varepsilon, \quad (82)$$

so y is the noisy version of the innovation channel.

Indeed, directly from the noisy model,

$$Y := X_1 - \alpha X_0 = \beta A_1 + \sqrt{\Delta} Z_1 - \alpha(\beta A_0 + \sqrt{\Delta} Z_0) \quad (83)$$

$$= \beta(A_1 - \alpha A_0) + \sqrt{\Delta}(Z_1 - \alpha Z_0) \quad (84)$$

$$= \beta \varepsilon + \sqrt{\Delta}(Z_1 - \alpha Z_0). \quad (85)$$

This identity is conceptually central:

$$\boxed{Y \text{ contains no clean } A_0 \text{ contribution at all.}} \quad (86)$$

The non-Gaussian signal therefore lives naturally in the innovation direction.

However, Y is not independent of X_0 for finite t , because both contain the shared noise Z_0 . We can quantify the residual coupling by rewriting r in (x_0, y) coordinates:

$$r = x_1 - \nu - \rho x_0 \quad (87)$$

$$= y + \alpha x_0 - \nu - \rho x_0 \quad (88)$$

$$= y + (\alpha - \rho)x_0 - \nu. \quad (89)$$

Now compute

$$\alpha - \rho = \alpha - \alpha \frac{\beta^2 \sigma_0^2}{v} = \alpha \frac{v - \beta^2 \sigma_0^2}{v} = \alpha \frac{\Delta}{v}. \quad (90)$$

Hence, with

$$c := \frac{\alpha \Delta}{v}, \quad (91)$$

we obtain

$$r = y + c(x_0 - \beta \mu_0). \quad (92)$$

The exact joint density in the new variables becomes

$$\boxed{p_t(x_0, y) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x_0 - \beta \mu_0)^2}{2v}\right) h_t(y + c(x_0 - \beta \mu_0)).} \quad (93)$$

This formula exhibits the separation:

- x_0 : Gaussian global mode;
- y : innovation channel;
- $c(x_0 - \beta \mu_0)$: finite-time leakage induced by the shared OU noise.

At $t = 0$, we have $\Delta = 0$, hence $c = 0$, and the factorization becomes exact:

$$p_t(x_0, y) = \frac{1}{\sqrt{2\pi \sigma_0^2}} \exp\left(-\frac{(x_0 - \mu_0)^2}{2\sigma_0^2}\right) \cdot \frac{1}{2b} \exp\left(-\frac{|y|}{b}\right). \quad (94)$$

So the clean model separates perfectly into a Gaussian site mode and a Laplace innovation mode.

10 Fourier representation and where linearity survives

10.1 Clean characteristic function

The clean characteristic function is

$$\hat{p}_0(k_0, k_1) := \mathbb{E} [\exp(i(k_0 A_0 + k_1 A_1))] \quad (95)$$

$$= \mathbb{E} [\exp(i(k_0 + \alpha k_1) A_0)] \mathbb{E} [\exp(ik_1 \varepsilon)]. \quad (96)$$

The first factor is Gaussian:

$$\mathbb{E} [\exp(i(k_0 + \alpha k_1) A_0)] = \exp\left(i\mu_0(k_0 + \alpha k_1) - \frac{\sigma_0^2}{2}(k_0 + \alpha k_1)^2\right). \quad (97)$$

The second factor is the Laplace characteristic function:

$$\mathbb{E} [\exp(ik_1 \varepsilon)] = \int_{\mathbb{R}} \frac{1}{2b} \exp\left(-\frac{|u|}{b}\right) \exp(ik_1 u) du = \frac{1}{1 + b^2 k_1^2}. \quad (98)$$

Therefore

$$\boxed{\hat{p}_0(k_0, k_1) = \exp\left(i\mu_0(k_0 + \alpha k_1) - \frac{\sigma_0^2}{2}(k_0 + \alpha k_1)^2\right) \frac{1}{1 + b^2 k_1^2}.} \quad (99)$$

The separation is again visible:

- quadratic exponential = Gaussian sector;
- rational factor $(1 + b^2 k_1^2)^{-1}$ = non-Gaussian innovation sector.

10.2 OU corruption in Fourier space

If $X = \beta A + \sqrt{\Delta} Z$, then characteristic functions satisfy

$$\hat{p}_t(k) = \exp\left(-\frac{\Delta}{2} \|k\|^2\right) \hat{p}_0(\beta k). \quad (100)$$

Thus

$$\boxed{\hat{p}_t(k_0, k_1) = \exp\left(i\beta\mu_0(k_0 + \alpha k_1) - \frac{\beta^2\sigma_0^2}{2}(k_0 + \alpha k_1)^2 - \frac{\Delta}{2}(k_0^2 + k_1^2)\right) \frac{1}{1 + b^2 k_1^2}.} \quad (101)$$

The OU channel multiplies the clean characteristic function by a Gaussian damping factor but does not remove the Laplace rational factor at finite time. This is the precise Fourier reason why finite-time smoothing does not Gaussianize the law.

10.3 Fourier representation in (x_0, y) coordinates

Let (k, ℓ) be dual variables to (x_0, y) . Since $x_1 = y + \alpha x_0$, the phase becomes

$$i(kx_0 + \ell y) = i((k - \alpha\ell)x_0 + \ell x_1). \quad (102)$$

Therefore

$$\hat{q}_t(k, \ell) = \hat{p}_t(k - \alpha\ell, \ell). \quad (103)$$

Substituting (101) yields

$$\hat{q}_t(k, \ell) = \exp\left(i\beta\mu_0 k - \frac{\beta^2\sigma_0^2}{2}k^2 - \frac{\Delta}{2}((k - \alpha\ell)^2 + \ell^2)\right) \frac{1}{1 + \tilde{b}^2\ell^2}. \quad (104)$$

This is the cleanest spectral separation of the model:

- k is the root/global mode;
- ℓ is the innovation mode;
- the only non-Gaussian factor is $(1 + \tilde{b}^2\ell^2)^{-1}$, purely along the innovation channel;
- the coupling between k and ℓ is entirely Gaussian and comes from the shared noise structure.

10.4 Linear equation for the residual density

The residual characteristic function is

$$\hat{h}_t(\ell) = \mathbb{E}[\exp(i\ell R_t)] = \frac{\exp(-\tau^2\ell^2/2)}{1 + \tilde{b}^2\ell^2}. \quad (105)$$

Hence

$$(1 + \tilde{b}^2\ell^2)\hat{h}_t(\ell) = \exp(-\tau^2\ell^2/2). \quad (106)$$

Taking inverse Fourier transforms gives the linear ordinary differential equation

$$(1 - \tilde{b}^2\partial_r^2)h_t(r) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{r^2}{2\tau^2}\right). \quad (107)$$

This equation is linear in h_t . The nonlinearity appears only when we pass to

$$\psi_t(r) = \frac{h'_t(r)}{h_t(r)}. \quad (108)$$

So the spectral problem is linear, whereas the score geometry in real space is nonlinear because it involves a logarithmic derivative.

11 Observed information, posterior covariance, and the meaning of the curvature field

Let

$$B := \beta A = (\beta A_0, \beta A_1), \quad X = B + \sqrt{\Delta}Z. \quad (109)$$

The Gaussian channel identity gives

$$\nabla_x \log p_t(x) = -\frac{x - \mathbb{E}[B | X = x]}{\Delta}. \quad (110)$$

Differentiate again. Using the Jacobian of the posterior mean, one obtains

$$\boxed{\mathcal{H}_t(x) = \frac{1}{\Delta} I - \frac{1}{\Delta^2} \text{Cov}(B | X = x).} \quad (111)$$

This identity is exact and very informative.

- The matrix $\mathcal{H}_t(x)$ is not literally a latent-space posterior precision.
- It is a data-space observed information matrix.
- It equals noise precision minus a correction coming from residual posterior uncertainty about the clean signal.

Averaging over X , one recovers the Fisher information matrix:

$$\mathbb{E}[\mathcal{H}_t(X)] = \mathbb{E}[\nabla \log p_t(X) \nabla \log p_t(X)^\top]. \quad (112)$$

So $\mathcal{H}_t(x)$ is the *local observed information*, whereas the Fisher information is its global average.

12 Why constant Hessian is equivalent to Gaussianity

Suppose $\mathcal{H}_t(x) \equiv Q$ is constant. Then

$$\nabla_x^2 \log p_t(x) = -Q. \quad (113)$$

Integrating twice yields

$$\log p_t(x) = -\frac{1}{2} x^\top Q x + \ell^\top x + c \quad (114)$$

for some vector ℓ and scalar c . Therefore p_t is Gaussian. Conversely, every Gaussian density has quadratic log-density and therefore constant Hessian. Hence

$$\boxed{\mathcal{H}_t(x) \text{ is constant} \iff p_t \text{ is Gaussian.}} \quad (115)$$

In the present model, the residual characteristic function is

$$\widehat{h}_t(\ell) = \frac{\exp(-\tau^2 \ell^2 / 2)}{1 + \tilde{b}^2 \ell^2}. \quad (116)$$

For any finite t , one has $\tilde{b} = \beta b > 0$. Therefore the fourth cumulant remains nonzero:

$$\kappa_4(R_t) = 12\tilde{b}^4 = 12\beta^4 b^4 \neq 0. \quad (117)$$

So the residual law is not Gaussian, the joint law is not Gaussian, and the Hessian cannot be constant at finite time.

Only in the limit $t \rightarrow \infty$ do we have $\beta \rightarrow 0$, $\tilde{b} \rightarrow 0$, and all non-Gaussian cumulants vanish.

13 Gaussian benchmark and the exact deviation from constant precision

To build a Gaussian benchmark with the same clean innovation variance, replace

$$\varepsilon \sim \text{Laplace}(0, b) \quad \text{by} \quad \varepsilon_G \sim \mathcal{N}(0, 2b^2), \quad (118)$$

since a centered Laplace variable with scale b has variance $2b^2$.

Then the same conditioning steps go through, but now the residual is Gaussian:

$$R_t^{(G)} \sim \mathcal{N}(0, \tau_G^2), \quad \tau_G^2 := \tau^2 + 2\tilde{b}^2. \quad (119)$$

Hence the Gaussian benchmark density is

$$p_t^{(G)}(x_0, x_1) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x_0 - \beta\mu_0)^2}{2v}\right) \cdot \frac{1}{\sqrt{2\pi\tau_G^2}} \exp\left(-\frac{r^2}{2\tau_G^2}\right). \quad (120)$$

Its score is linear:

$$\mathcal{S}^{(G)}(x, t) = \begin{pmatrix} -\frac{x_0 - \beta\mu_0}{v} + \frac{\rho r}{\tau_G^2} \\ -\frac{r}{\tau_G^2} \end{pmatrix}, \quad (121)$$

and its precision matrix is constant:

$$Q_t = \begin{pmatrix} \frac{1}{v} + \frac{\rho^2}{\tau_G^2} & -\frac{\rho}{\tau_G^2} \\ -\frac{\rho}{\tau_G^2} & \frac{1}{\tau_G^2} \end{pmatrix}. \quad (122)$$

Compare this with (75). The difference is

$$\Delta\mathcal{H}_t(x) := \mathcal{H}_t(x) - Q_t = \left(\kappa_t(r) - \frac{1}{\tau_G^2} \right) \begin{pmatrix} \rho^2 & -\rho \\ -\rho & 1 \end{pmatrix}. \quad (123)$$

So every deviation from the Gaussian benchmark is concentrated in the single scalar difference

$$\kappa_t(r) - \frac{1}{\tau_G^2}. \quad (124)$$

Interpretation:

- near $r = 0$, the Laplace model is sharper than a Gaussian fit, so curvature is larger;
- in the tails, the Laplace model is heavier, so curvature softens and tends to zero;
- the Gaussian model cannot reproduce this center-versus-tail stiffness change because its curvature is constant.

14 Interpretation of the curvature heatmaps

Figure 1 is the visual manifestation of (76). Since

$$H_{11}(x) = \kappa_t(r), \quad H_{01}(x) = -\rho\kappa_t(r), \quad r = x_1 - \nu - \rho x_0, \quad (125)$$

all patterns in the heatmaps are transported copies of the one-dimensional profile κ_t .

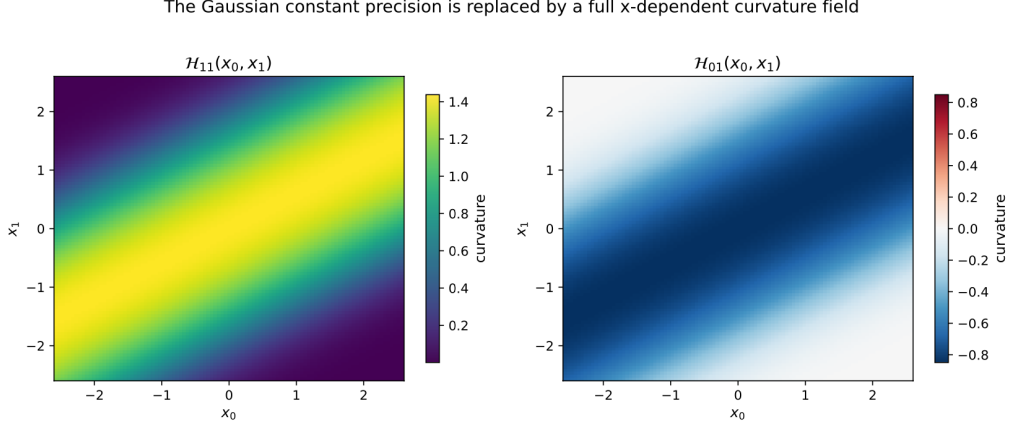


Figure 4: **Two-dimensional curvature field.** Heatmaps of the exact entries $(\mathcal{H}_t(x_0, x_1))_{11}$ and $(\mathcal{H}_t(x_0, x_1))_{01}$. The Gaussian notion of one fixed precision matrix is replaced by an x -dependent field.

Figure 1: Two entries of the exact curvature field. The visible two-dimensional pattern is entirely generated by the one-dimensional residual variable $r = x_1 - \nu - \rho x_0$. Curvature is concentrated near the residual band $r \approx 0$, which corresponds approximately to the innovation line $x_1 \approx \alpha x_0$ when t is small or moderate.

14.1 Why the diagonal band appears

Curvature concentrates where the residual density is most sharply curved, namely around $r = 0$. Therefore the high-curvature region is the straight band

$$x_1 \approx \nu + \rho x_0. \quad (126)$$

If $\mu_0 = 0$ and Δ is not too large, then

$$\rho = \alpha \frac{\beta^2 \sigma_0^2}{\beta^2 \sigma_0^2 + \Delta} \approx \alpha, \quad (127)$$

so the band appears visually near the clean innovation line $x_1 \approx \alpha x_0$.

14.2 Why this is the smoothed remnant of a cusp

At time $t = 0$, the clean free energy is

$$F_0(x_0, x_1) = \frac{(x_0 - \mu_0)^2}{2\sigma_0^2} + \frac{|x_1 - \alpha x_0|}{b} + \text{const.} \quad (128)$$

The second derivative of $|u|$ is distributional:

$$\frac{d^2}{du^2}|u| = 2\delta(u). \quad (129)$$

Therefore the clean curvature is a singular sheet concentrated on the line $x_1 = \alpha x_0$:

$$\mathcal{H}_0(x) = \begin{pmatrix} \sigma_0^{-2} & 0 \\ 0 & 0 \end{pmatrix} + \frac{2}{b}\delta(x_1 - \alpha x_0) \begin{pmatrix} \alpha^2 & -\alpha \\ -\alpha & 1 \end{pmatrix}. \quad (130)$$

Finite OU smoothing replaces this singular delta-sheet by the smooth diagonal band seen in the figure.

14.3 Why the off-diagonal entry is negative

When $\alpha > 0$, one has $\rho > 0$. Moreover the residual density h_t is log-concave because it is the convolution of two log-concave densities (Gaussian and Laplace). Therefore

$$\kappa_t(r) = -\frac{d^2}{dr^2} \log h_t(r) \geq 0. \quad (131)$$

Hence

$$H_{01}(x) = -\rho \kappa_t(r) \leq 0. \quad (132)$$

So the off-diagonal field is expected to be negative in the band and to relax toward 0 away from it.

15 Direct answers to the main conceptual questions

15.1 Why finite-time OU smoothing does not Gaussianize Laplace

The reason is exact in Fourier space:

$$\hat{h}_t(\ell) = \frac{\exp(-\tau^2 \ell^2 / 2)}{1 + \tilde{b}^2 \ell^2}. \quad (133)$$

The Gaussian damping factor smooths high frequencies, but the rational Laplace factor remains present for every finite t because $\tilde{b} = \beta b > 0$. Therefore higher cumulants do not vanish at finite time. In particular,

$$\kappa_4(R_t) = 12\beta^4 b^4 \neq 0. \quad (134)$$

No exact Gaussianization occurs unless $t \rightarrow \infty$.

15.2 Is the curvature field a local Fisher information?

Pointwise, $\mathcal{H}_t(x)$ is best understood as the Hessian of the free energy or local observed information. After averaging over x , it becomes Fisher information:

$$\mathbb{E}[\mathcal{H}_t(X)] = \mathbb{E} \left[\nabla \log p_t(X) \nabla \log p_t(X)^\top \right]. \quad (135)$$

So the answer is:

$$\text{pointwise: observed information,} \quad \text{in expectation: Fisher information.} \quad (136)$$

15.3 Is $\mathcal{H}_t(x)$ the posterior precision of $A | X = x$?

Not literally. The exact identity is

$$\mathcal{H}_t(x) = \frac{1}{\Delta} I - \frac{1}{\Delta^2} \text{Cov}(\beta A | X = x). \quad (137)$$

So $\mathcal{H}_t(x)$ is controlled by posterior covariance, not by a latent posterior precision matrix. In the Gaussian case the distinction is less important because Gaussian posteriors are fully described by covariance/precision. In the Laplace case, the posterior is not Gaussian and no single precision matrix captures the whole posterior shape.

16 Modification guide: what changes when you change the model?

This section is included so the note can be edited and repurposed easily.

16.1 Changing the clean innovation law

If the Laplace innovation is replaced by another centered law with characteristic function $\hat{f}_\varepsilon(\ell)$, then the whole derivation changes only in one place:

$$\hat{h}_t(\ell) = \exp(-\tau^2 \ell^2 / 2) \hat{f}_\varepsilon(\beta \ell). \quad (138)$$

Everything else still follows from

$$p_t(x_0, x_1) = \text{Gaussian in } x_0 \times h_t(x_1 - \nu - \rho x_0). \quad (139)$$

So for $K = 1$, many non-Gaussian models remain one-dimensional after conditioning and residualization.

16.2 Changing α

The parameter α affects

$$\rho = \alpha \frac{\beta^2 \sigma_0^2}{v}, \quad \nu = \alpha \beta \frac{\Delta \mu_0}{v}, \quad \tau^2 = \Delta + \alpha^2 \beta^2 \frac{\sigma_0^2 \Delta}{v}. \quad (140)$$

So increasing $|\alpha|$ changes both the orientation of the residual band and the amount of Gaussian broadening in the innovation channel.

16.3 Changing the prior for A_0

The present exact Gaussian formulas for $m(x_0)$ and s^2 rely on the Gaussian prior for A_0 . If that prior changes, then the posterior $A_0 | X_0$ may cease to be Gaussian, and the residual

representation must be modified. In that case, the exact one-dimensional closure may fail, even though the Markov decomposition still gives a local structure.

17 Compact summary of the main formulas

For quick reference, here are the main formulas in one place.

Structural constants

$$\beta = e^{-t}, \quad \Delta = 1 - \beta^2, \quad v = \beta^2 \sigma_0^2 + \Delta, \quad (141)$$

$$\tilde{b} = \beta b, \quad s^2 = \frac{\sigma_0^2 \Delta}{v}, \quad \tau^2 = \Delta + \alpha^2 \beta^2 \frac{\sigma_0^2 \Delta}{v}, \quad (142)$$

$$\rho = \alpha \frac{\beta^2 \sigma_0^2}{v}, \quad \nu = \alpha \beta \frac{\Delta \mu_0}{v}, \quad c = \frac{\alpha \Delta}{v}. \quad (143)$$

Residual variable

$$r = x_1 - \nu - \rho x_0 = y + c(x_0 - \beta \mu_0), \quad y = x_1 - \alpha x_0. \quad (144)$$

Joint density

$$p_t(x_0, x_1) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x_0 - \beta \mu_0)^2}{2v}\right) h_t(r). \quad (145)$$

Normal–Laplace residual density

$$h_t(r) = \frac{\exp(a^2/2)}{2\tilde{b}} \left[\exp\left(-\frac{r}{\tilde{b}}\right) \Phi\left(\frac{r}{\tau} - a\right) + \exp\left(\frac{r}{\tilde{b}}\right) \Phi\left(-\frac{r}{\tau} - a\right) \right], \quad a = \frac{\tau}{\tilde{b}}. \quad (146)$$

Score

$$\mathcal{S}(x, t) = \begin{pmatrix} -\frac{x_0 - \beta \mu_0}{v} - \rho \psi_t(r) \\ \psi_t(r) \end{pmatrix}, \quad \psi_t(r) = \frac{h'_t(r)}{h_t(r)}. \quad (147)$$

Curvature

$$\mathcal{H}_t(x) = \begin{pmatrix} \frac{1}{v} & 0 \\ 0 & 0 \end{pmatrix} + \kappa_t(r) \begin{pmatrix} \rho^2 & -\rho \\ -\rho & 1 \end{pmatrix}, \quad \kappa_t(r) = -\frac{d^2}{dr^2} \log h_t(r). \quad (148)$$

Fourier representation

$$\hat{p}_t(k_0, k_1) = \exp\left(i\beta\mu_0(k_0 + \alpha k_1) - \frac{\beta^2 \sigma_0^2}{2}(k_0 + \alpha k_1)^2 - \frac{\Delta}{2}(k_0^2 + k_1^2)\right) \frac{1}{1 + \tilde{b}^2 k_1^2}. \quad (149)$$

18 Closing interpretation

For $K = 1$, the Laplace AR(1) model under OU corruption is not an arbitrary two-dimensional non-Gaussian problem. It is a one-dimensional non-Gaussian bond attached to a Gaussian site mode. All nonlinear score effects, all curvature variations, and all deviations from the Gaussian benchmark are carried by the single residual coordinate

$$r = x_1 - \nu - \rho x_0. \tag{150}$$

That is why the heatmaps are so structured: they are not generic two-dimensional landscapes but one-dimensional profiles lifted along straight residual bands. This is the precise sense in which Markovianity creates locality and Laplace innovations create nonlinear bond geometry.